

ORGANIC CHEMISTRY – ADDITION POLYMERISATION

KEY VOCABULARY

Addition polymerisation: alkene monomers join with no other product.

Monomer: the small unsaturated molecule (has a C=C).

Repeat unit: the part of the polymer in brackets, with n.

Poly(ethene): the polymer made from ethene monomers.

COMMON MISCONCEPTIONS

~~— Addition polymerisation makes water too.~~

✓ There is only ONE product — the polymer.

~~— Saturated monomers can polymerise this way.~~

✓ Only unsaturated (C=C) monomers can.

~~— The repeat unit keeps the double bond.~~

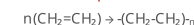
✓ The C=C opens to single bonds in the polymer.

MONOMER → POLYMER

ADDITION POLYMERISATION



The C=C double bond opens to form single bonds



1 DEFINE

State what is meant by **addition polymerisation**. [1 mark]

2 MONOMER NEEDED

Explain why only alkenes can undergo addition polymerisation. [2 marks]

3 DRAW REPEAT UNIT

Draw the **repeat unit** of poly(ethene). [2 marks]

4 NAME POLYMER

Name the polymer made from propene monomers. [1 mark]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Addition polymerisation makes ____ product as the ____ bonds open up.

one

double

two

single

6 ENVIRONMENT

Explain why poly(ethene) waste is a problem for the environment. [2 marks]